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Task Description:	Most major cereal grain crops are monocots. Yet, most investigations of plant adaptation to the spaceflight environment have been carried out on the dicotyledonous model plant Arabidopsis thaliana. It remains unknown whether the conclusions from such studies can be extrapolated to monocotyledonous plants. To fill this knowledge gap, we propose to investigate the growth, development, and transcriptome profiles of Brachypodium distachyon seedlings under spaceflight conditions, and compare these adaptive responses to those recently documented for Arabidopsis seedlings. These investigations will lead to a better understanding of the molecular and developmental mechanisms that contribute to Brachypodium adaptation to spaceflight conditions. In the long term, these data may also lead to the development of better mitigation strategies aimed at improving monocot adaptability to spaceflight parameters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Because most cultivated crops are monocotyledonous plants, there is tremendous interest in better understanding the molecular mechanisms that govern their growth behaviors and adaptation to stress. In addition to contributing to a better understanding of the molecular mechanisms that modulate plant adaptation to the microgravity environment, an important step toward engineering crops that are better suited to serve as components of the bioregenerated life-support systems during spaceflight, our work with Brachypodium distachyon should provide new insights into the molecular mechanisms that control root growth and development. Because root growth and its architecture have recently been recognized as key contributors to plant productivity and adaptation to environmental stress, our research is also likely to have long-term impact in agriculture, horticulture, and forestry. Furthermore, many of the behaviors under investigation involve auxin-related processes. This plant hormone is a key contributor to most aspects of plant growth, development, and response to the environment. Therefore, this project is likely to yield important new insights into some of the key mechanisms that control plant growth, development, and response to biotic and abiotic stresses.
	FINAL REPORTING NOVEMBER 2019: Most of the studies of plant growth and adaptation to the microgravity environment have been carried out using Arabidopsis thaliana, a model dicot plant. However, most of the cultivated crops on Earth are monocots. Therefore, the purpose of our project was to investigate the impact of the microgravity environment present on the International Space Station (ISS) on the germination, growth, and morphology of Brachypodium distachyon seedlings, and evaluate their molecular adaptation to this environment using RNAseq analyses of transcription profiles. To do this, we first developed a new foam-based plant growth unit (named the APEX Growth Chamber), in collaboration with Dr. Howard Levine and the Kennedy Space Center Flight Support Team. This chamber allows planting Brachypodium seeds within a block of foam on the ground, before flight. The seeds remain dry during take off and travel to the ISS. Upon arrival, an astronaut (Dr. Scott Tingle in our experiment) injects a nutritive solution into the foam, triggering seed germination and growth with the VEGGIE growth unit. Using this approach, we were able to grow 3 accessions of Brachypodium distachyon (Bd21, BD21-3, and Gaz-8) on ISS with four biological repeats per accession (four APEX Growth Chambers containing 24 seeds each). We also carried out two ground controls. One of these ground controls included germinating and growing the same accessions on the ground under conditions that mimicked those encountered on ISS, with a 48-h delay (ground control 1). A second ground control involved growing Bd21 seedlings under the same conditions for three days, followed by 5-min gravistimulation (GS) or 5-min mechanostimulation (MS) (ground control 2). At the end of each experiment, the seedlings were photographed, and the plant material was harvested and fixed in RNALater in KFT fixation units for subsequent analyses. Images were used to quantify root and shoot growth under both conditions, whereas plant materials were dissected to separate shoots and roots for subsequent expression profiling. RNA was extracted for RNASeq analysis. Our results show that the plants grown under microgravity displayed shorter, very hairy roots. Shoots were shorter in only one of the accessions tested (Bd21). Expression profiling revealed a large number of Bd21 genes whose expression either increased or decreased in shoots and/or roots under microgravity. Most of the differentially expressed genes were organ-specific, and did not respond to either GS or MS under ground-control conditions. The lists of microgravity-response genes were enriched for genes predicted to function in plant responses to environmental and oxidative stress as well as reactive oxygen species. They were also enriched for genes involved in radiation response, implying that Brachypodium might serve as a good model to investigate the effects of cosmic radiation on plants. Few of the microgravity-response genes were also differentially expressed in response to GS or MS on the ground. Those that did respond to both GS and microgravity constitute excellent candidates for a function in gravity signal transduction. Overall, our studies identify several genes and pathways that could be further engineered to improve monocot plant adaptation to the microgravity environment, facilitating their use in bioregenerative life support systems for long-term space-exploration missions. Ground based studies were also carried out to investigate the natural variation that exists between Brachypodium accessions for root-growth and behavioral responses to GS. These studies led to the discovery of several key loci that may contribute to the regulation of proprioception in plants. Experimental Results and Conclusions Development of a plant growth chamber to test Brachypodium distachyon seedlings growth under microgravity conditions To test the ability of Brachypodium distachyon seedlings to germinate, grow, and adapt to the microgravity environment of the ISS, we developed a novel plant growth unit (named APEX Growth Chamber) in collaboration with Dr. Howard Levine and the NASA Kennedy Space Center Flight Support team. This system is based on an Oasis foam-based growth system previously used by Dr. Levine to grow plants in space. With this setup, Brachypodium seeds are inserted pre-flight into a dry block of foam that is surrounded by gauze and a nylon mesh, and is mounted on a medium injection device within a Magenta-box assembly. The seeds are kept in a dormant dry state, unable to germinate during spaceflight to the ISS. When the experiment is ready for activation on ISS, an astronaut can easily inject liquid growth medium into the foam support through the injection device, thereby imbibing the seeds and triggering their germination under red light. This novel plant germination and growth system had to be developed because early preparatory experiments had demonstrated that the far-red light pretreatment typically used to prevent the germination of Arabidopsis thaliana seedlings on agar-based media during takeoff and travel to the ISS, has an opposite effect on agar-medium-embedded Brachypodium seeds, triggering their germination. An alternative light-based inhibition protocol we developed to inhibit Brachypodium seed germination during spaceflight could not be implemented because it required continuous seed exposure to blue-light and cold, thereby requiring excessive amounts of energy during a sensitive period of

spaceflight. Our flight experiment and corresponding ground control involved a total of 15 APEX Growth Units per condition, including 12 for each experiment, and 3 spare units. Three distinct accessions of Brachypodium distachyon were used, including Bd21 (whose genome serves as a reference for Brachypodium), Bd21-3 (which has been optimized for use in transformation experiments), and Gaz-8 (which displays different root-growth behaviors relative to Bd21 and Bd21-3 when grown on the ground). Four chambers (biological repeats) per accession were tested under microgravity conditions on ISS, and another four were exposed to 1-g on the ground under conditions that mimicked the flight experiment (ground control).

This material was prepared for flight to the ISS on March 27-30, 2018, and launched on Space X-14 on April 2, 2018. Upon transferring the 12 seeded APEX growth chambers from stowage into the ISS, astronaut Scott Tingle activated the experiment by injecting growth medium into the units on April 12, 2018. Seeds were allowed to germinate in the presence of red light for 24 hours, then blue and green lights were turned on and the plants were allowed to grow for three more days. At the end of this growth period, astronaut Tingle collected each APEX growth unit, took photographs of the seedlings growing on each face of the foam block, harvested the seedlings and placed them into Kennedy Space Center Fixation Tubes (KFT) for fixation in the presence of RNAlater. For each APEX Growth Unit, a few seedlings displayed roots that grew away from the foam surface and showed evidence of stress. These seedlings were harvested separately from those whose root tips were still contacting the block surface at harvesting time. Fixation in RNAlater was carried out at room temperature for a period of 24 h. The KFTs were then transferred to a cold storage device (-80°C) and returned to the ground on the Dragon capsule.

The environmental conditions experienced by the plants during their growth in VEGGIE on ISS were recorded, and then recapitulated in plant growth units at KSC during a first ground-based control. At the end of this experiment, the seedlings were also fixed in RNAlater at room temperature for 24 hours, then frozen at -80°C. These ground-control samples were also returned to the Principal Investigator (PI) laboratory, as were the pictures of seedlings grown under both microgravity and 1-g conditions.

Microgravity-grown seedlings are morphologically altered compared to ground-controls The pictures of Brachypodium seedlings taken after 4 days of growth in the APEX Growth Units under microgravity and on the ground, were used to evaluate a potential effect of ISS-microgravity exposure on organs growth and seedling morphology. Seedlings of all three accessions displayed significantly shorter roots when grown under microgravity relative to ground controls. Furthermore, the shoots of microgravity-grown Bd21 seedlings were also shorter than ground controls, whereas those of Bd21-3 and Gaz-8 were similar under both conditions. We conclude that Brachypodium distachyon accessions display distinct adaptive responses to the microgravity environment encountered on ISS. In addition to accession-specific alterations of organs growth, we also observed that the primary roots of microgravity-grown seedlings displayed longer root hairs than those of ground-control samples, and this phenotype was observed for all three accessions tested. This phenotype is rather surprising as previous studies had shown an opposite effect of microgravity on Arabidopsis thaliana root hairs (4).

Brachypodium distachyon seedlings display organ-specific transcriptional responses to microgravity relative to ground control

To better understand the mechanisms that contribute to Brachypodium seedling adaptation to the microgravity environment of ISS, root and shoot tissues were dissected from frozen fixed Bd21 seedlings that had been exposed to the microgravity environment on ISS as well as ground-control materials. Total RNAs were extracted from these tissues and used to build TruSeq Stranded Total RNA libraries, which were sequenced in a Novaseq sequencer programmed to generate 150-bp paired-end reads. Library sequencing generated an average of 25 to 30 million reads per sample. Reads were trimmed to remove the adapter sequences using SKEWER, then filtered with FASTX_quality_filter to ensure that only high-quality reads (Phred scores above 35) are used for sequence alignments. The quality-filtered reads were mapped on the Brachypodium reference genome (version 3.1, from Phytozome) using the Bowtie2 algorithm. The TopHat 2 splice-junction mapper was used to compile transcript contigs, paying close attention to potential splice variants. HTseq-count was then used to define the numbers of reads per kilobase of transcript per million mapped reads (RPKM) for genes and isoforms in each analyzed sample. DESeq (1) was used to normalize the numbers of reads and identify significantly differentially expressed transcripts. EdgeR (9) was also used as an alternative method to identify differentially expressed genes, and genes identified as significantly differentially expressed by both packages were retained for further analysis. The results of our DESeq comparisons are presented below.

A principle component analysis (PCA) of expression profiles between experimental repeats revealed separate clustering between ground-control and microgravity-exposed samples, suggesting that there is more variation in expression profiles between environmental conditions than there is between repeats within a condition. Furthermore, ground-control root samples grouped more tightly in a sub-region of the PCA graph than those from microgravity-exposed root samples, suggesting a larger variation in expression profiles between repeats of the microgravity-exposed root samples relative to the ground controls.

Analysis of differential expression between microgravity-exposed and ground control seedlings identified 661 and 267 up-regulated genes in microgravity-exposed shoots and roots, respectively, relative to ground controls. Additionally, 1015 and 505 genes were found to be down-regulated in microgravity-exposed shoots and roots, respectively. A Gene Ontology (GO) annotation of these differentially expressed genes revealed an over-representation of genes potentially involved in plant responses to environmental stimuli, oxidative stress, reactive oxygen species and radiation, amongst other things.

Even though many genes ended up being differentially expressed between microgravity and ground control samples, most were organ-specific. This is consistent with similar observations made with Arabidopsis thaliana seedlings (5; 7). In fact, only 90 of the significantly differentially expressed genes are shared between roots and shoots. Of these, 35 are up-regulated and 30 down-regulated in both roots and shoots under microgravity. These genes constitute excellent candidates for a contribution to plant adaptive response to the microgravity environment of ISS. The remaining genes display opposite responses between shoots and roots. Several investigations of Arabidopsis seedling adaptation to spaceflight have also uncovered evidence of transcriptional responses involving genes that contribute to environmental and oxidative stress responses, as well as responses to reactive oxygen species (2-4; 6-8). On the other hand, our observation of enhanced expression of genes involved in radiation response under microgravity on ISS is particularly interesting considering the risks of exposure to cosmic radiations encountered by living organisms during spaceflight.

A limited number of Brachypodium distachyon genes found to respond to the microgravity environment of ISS also respond to GS and/or MS on the ground.

Task Progress:

Because Brachypodium distachyon seedling-organ responses to GS had not been previously characterized, we carried out a second ground-control experiment in this project, aimed at identifying genes whose expression is significantly altered in response to GS and/or MS. We germinated and grew Brachypodium Bd21-accession seeds in APEX Growth Chambers, within a VEGGIE growth unit at KSC under conditions that were programmed to mimic those recorded on ISS during the spaceflight experiment. After 4 days of growth, the APEX growth units were either gravity-stimulated by 180° reorientation, mechano-stimulated by 360° rotation, or not stimulated (control), and the seedlings were allowed to grow in their new orientation for an additional 5 min. Each treatment involved four repeats (four APEX Growth units). After stimulation, the seedlings were harvested and fixed in RNAlater (7/19/2018), then frozen at -80°C after 1 day at room temperature. These samples were returned to the University of Wisconsin-Madison, and dissected to separate shoot and root tissues. RNA was extracted from each sample and analyzed by RNAseq to identify genes that are differentially expressed (activated or repressed) in response to GS or MS. Similar sequence-analysis packages were used as those described above for RNASeq analysis of Brachypodium organ responses to microgravity.

This analysis revealed only small numbers of significantly differentially expressed genes between GS-/MS-induced samples and ground controls. It is, however, particularly compelling to note that 10 genes were found to be differentially expressed in response to both GS and microgravity exposure in roots. Six of these genes were repressed upon GS and activated upon exposure to microgravity, whereas the other four genes were activated in response to GS and repressed by exposure to microgravity. Therefore, these 10 genes constitute strong candidates for a contribution to gravity response in roots. They will be subjected to functional characterization.

Equally interesting is another list of 10 genes that are transcriptionally responsive to both MS and microgravity exposure. Of these, eight were found to be down-regulated in response to MS and up-regulated upon exposure to microgravity, one was up-regulated in response to MS and down-regulated in response to microgravity, and the last one was found to be up-regulated in response to both treatments. One of these genes was also found in the list of 10 genes that are responsive to both GS and microgravity exposure. Overall, the latter group of 10 genes jointly regulated by MS and microgravity exposure are likely to contribute to plant responses to their mechanical environment. They will also be prioritized for further functional characterization.

In conclusion, we note that a vast majority of the genes found to be differentially expressed upon GS or MS on the ground are not transcriptionally responsive to the microgravity conditions experienced on ISS, suggesting that Brachypodium seedling organs interpret the microgravity environment of ISS very differently from GS and/or MS on the ground. This observation holds true for both shoot and root samples. It is compatible with similar observations made by others using Arabidopsis thaliana as a model (2-4; 6-8). A manuscript describing this work is in preparation.

Using a Genome-Wide Association Study approach to identify loci that contribute to root gravitropism and oscillatory root growth behavior in Brachypodium distachyon seedlings

In addition to the APEX-06 experiment described above, we have also completed our comparative analysis of root gravitropism and oscillatory root growth behavior on agar surfaces for 46 distinct Brachypodium distachyon accessions, using time lapse-analysis of root-growth behavior coupled with automated computer-driven biometric analysis of imaged roots. Our quantitative analysis of root growth behavior included determinations of root growth rates, root tip angles from the vertical, and elemental curvatures along the roots.

When growing on vertical plates, Brachypodium roots from most accessions grow straight down, as dictated by gravitropism. However, while doing so, the root-tip also tends to curve leftward and rightward of the vertical. The tip curvatures associated with these oscillations typically disappear as the differentially elongated cells migrate from the distal side of the elongation zone to more proximal regions, an auto-straightening process that is reminiscent of proprioception in animal systems. Upon 90-degree reorientation within the gravity field (gravity stimulation), the roots initiate a phase of rapid downward bending, which is followed by a second, slower phase of curvature response that is accompanied by root-tip oscillations similar to those preceding the gravity stimulus. Distinct Brachypodium accessions differ from each other in the frequency and amplitude of their root tip oscillations, as well as in the bending rate following gravity stimulation.

To better capture the genetic basis of variation between accessions for these traits, we developed a mathematical model that recapitulates the evolution of root tip angles from the vertical before, during and after gravity stimulation. When applied to the 46 accessions under investigation, this model is very effective at recapitulating accession-specific growth behaviors, with best-fit curves correlating well with the observed data (R-square larger than 0.9). These best-fit equations, along with Fast Fourier Transform analysis, allowed estimation of descriptive parameters for these behaviors, including root growth rate, rapid bending rates during phase 1 of gravitropism, and frequency and amplitude of oscillations.

Because each one of these 46 accessions had previously been subjected to genome sequencing, we were able to use genome-wide association studies (GWAS) to identify single nucleotide polymorphisms (SNPs) that are associated with these parameters. This analysis allowed us to identify linked loci whose contribution to oscillatory root growth behavior are being investigated using a combination of expression analysis, reverse genetics and cell biology approaches. A paper describing this research is also being prepared for publication.

Broader Impact of Our Work

Overall, these studies may impact our understanding of the molecular mechanisms that govern the regulation of coordinated cell expansion and its roles in growth regulation, plant morphogenesis and plant organs' growth responses and adaptation to gravity and/or microgravity. In the longer term, our studies may enable us to engineer plants that are better suited to function as key components of bioregenerative life support systems for long-term space exploration missions, or more simply for effective growth and productivity in natural and/or agricultural ecosystems on Earth.

New Technology Development: The foam-based vessel (APEX Growth Chamber) we developed in collaboration with Dr. Howard Levine and the KSC Flight Support Team has been submitted to NASA Technology Transfer System (KSC-14282).

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JUNE 2019 REPORT: As summarized in our previous progress reports, we developed a novel plant growth unit (named APEX growth chamber) to support the growth of Brachypodium distachyon seedlings under microgravity in the International Space Station (ISS). This system is based on an Oasis foam-based growth device previously used by Dr. Howard Levine to grow plants in space (NASA Space Biology Program). Our units were designed in a collaborative effort involving Dr. Levine and the NASA Flight Support Team. With this system, seeds are inserted pre-flight into a dry block of foam, which is mounted on a medium injection device within a Magenta box. The seeds are kept in a dormant dry state, unable to germinate during spaceflight to the ISS. When the experiment is ready for initiation on ISS, an astronaut can easily inject liquid growth medium into the foam support through the injection device, thereby imbibing the seeds and triggering their germination under red light.

The flight experiment and corresponding ground control were carried out in March 2018 (APEX-06 mission). These experiments involved a total of 15 APEX Growth Units per condition, including 12 for each experiment and 3 spare ones. Three distinct accessions of Brachypodium distachyon were used, including Bd21 (whose genome serves as a reference for Brachypodium), BD21-3 (which has been optimized for use in transformation experiments), and Gaz-8 (which displays different root growth behaviors relative to Bd21 and Bd21-3 when grown on the ground). Four chambers (biological repeats) per accession were tested under microgravity conditions on ISS, and another four were exposed to 1-g on the ground under conditions that mimicked the flight experiment (ground control).

As previously reported, this material was prepared for flight to the ISS on March 27-30, 2018, and launched on SpaceX-14 on April 2, 2018. Upon transferring the 12 seeded APEX growth chambers from stowage into the ISS, astronaut Scott Tingle activated the experiment by injecting growth medium into the units on April 12, 2018. Seeds were allowed to germinate in the presence of red light for 24 hours, then blue and green lights were turned on and the plants were allowed to grow for three more days. At the end of this growth period, astronaut Tingle collected each APEX growth unit, took photographs of each face, harvested the seedlings, and placed them into KFT for fixation in the presence of RNAlater. For each APEX Growth Unit, a few seedlings displayed roots that grew away from the foam surface and showed evidence of stress. These seedlings were harvested separately from those whose root tips were still contacting the block surface at harvesting time. Fixation in RNAlater was carried out at room temperature for a period of 24 h. The KFTs were then transferred to a cold storage device (-80°C) and returned to the ground on the Dragon capsule.

The environmental conditions experienced by the plants during their growth period in VEGGIE on ISS were recorded, and then recapitulated in plant growth units at KSC during a first ground-based control. At the end of this experiment, the seedlings were also fixed in RNAlater. After similar storage under -80°C conditions, the samples were returned to the Principal Investigator (PI) laboratory, as were the pictures of seedlings grown under both microgravity and 1-g conditions. As reported last year, seedlings grown under microgravity displayed significant root-growth inhibition compared to ground controls for all accessions tested. The shoots of microgravity-grown Bd21 seedlings were also shorter than ground control, but those of Bd21-3 and Gaz-8 were similar under both gravity conditions. We conclude that distinct Brachypodium distachyon accessions display distinct adaptive responses to the microgravity environment encountered on ISS. In addition to accession-specific slight alterations of organs growth, we also observed that the primary roots of microgravity-grown seedlings displayed longer root hairs than those of ground-control samples.

To better understand the mechanisms that contribute to Brachypodium seedlings adaptation to the microgravity environment of ISS, root and shoot tissues were dissected from frozen fixed Bd21 seedlings that had been exposed to the microgravity environment on ISS as well as ground-control material. Total RNAs were extracted from these tissues and used to build TruSeq Stranded Total RNA libraries, which were sequenced in a Novaseq sequencer programmed to generate 150-bp paired-end reads. Library sequencing generated an average of 25 to 30 million reads per sample. Sequence analysis involved a pipeline that included FastQC to control for the quality of sequence reads and filter out poor reads, Tophat2 to align each reads to the reference Brachypodium genome, Bowtie2 to build an index, HTseq to count the number of reads assigned to each annotated gene/transcript, and DEseq (R package) to evaluate differential expression between microgravity-exposed and ground control samples.

A principle component analysis (PCA) of expression profiles between experimental repeats revealed separate clustering between ground-control and microgravity-exposed samples, suggesting that there is more variation in expression profiles between environmental conditions than there is between repeats within a condition. Furthermore, ground-control root samples grouped more tightly in a sub-region of the PCA graph than those from microgravity-exposed root samples, suggesting a larger variation in expression profiles exists between repeats of the microgravity-exposed root samples relative to the ground controls.

Initial analysis of differential expression between microgravity-exposed and ground control seedlings identified 398 and 159 significantly up-regulated genes in microgravity-exposed shoots and roots, respectively, relative to ground controls. Additionally, 867 and 380 significantly down-regulated genes were found in microgravity-exposed shoots and roots, respectively. A Gene Ontology (GO) annotation of these differentially expressed genes revealed an over-representation of genes potentially involved in plant responses to environmental stimuli and oxidative stress, amongst other things.

Because the Brachypodium distachyon seedling organ responses to gravity stimulation have not been well characterized, we also carried out a second control experiment in this project, aimed at identifying genes whose expression is significantly altered in response to gravity- and/or mechano-stimulation. As reported last year, we germinated and grew Bd21 seeds in APEX Growth Chambers, within a VEGGIE unit, under growth conditions that were programmed to mimic those recorded on ISS during the spaceflight experiment. After four days of growth, the APEX growth units were either gravity-stimulated by 180° reorientation, mechano-stimulated by 360° rotation, or unstimulated (control), and the seedlings were allowed to grow in their new position for five minutes. Each stimulus involved four repeats (four APEX Growth units). After 5-min incubations, the seedlings were harvested and fixed in RNAlater (7/19/2018), then frozen at -80°C after one day of fixation. These samples were returned to the University of Wisconsin-Madison where each sample was dissected to separate shoot and root tissues. RNA was extracted from each sample and analyzed by RNAseq to identify genes that are differentially expressed between treatments. Currently, the analytical pipeline described above is being used to analyze the RNAseq profiles and identify genes that are significantly differentially expressed (activated or repressed) in response to mechano- or gravity-stimulation. The groups of genes found to be differentially expressed under microgravity on ISS will be compared to those that are differentially expressed in response to gravity- or mechano-stimulation under 1-g. Reverse genetics will then be used to investigate the contribution of these genes to plant responses to microgravity, gravity- or mechano-stimulation.

In addition to the APEX-06 experiment described above, we have also completed our comparative analysis of root gravitropism and oscillatory root growth behavior on agar surfaces for 45 distinct Brachypodium distachyon accessions, using time lapse-analysis of root growth behavior coupled with automated computer-driven biometric analysis of imaged roots. Our quantitative analysis of root growth behavior included determinations of root growth rates, root tip angles from the vertical, and elemental curvatures along the roots.

When growing on vertical plates, Brachypodium roots from most accessions grow straight down, as dictated by gravitropism. However, while doing so, the root-tip also tends to curve leftward and rightward of the vertical. The tip curvatures associated with these oscillations typically disappear as the differentially elongated cells migrate from the distal side of the elongation zone to more proximal regions, an auto-straightening process that is reminiscent of proprioception in animal systems. Upon 90-degree reorientation within the gravity field (gravity stimulation), the roots initiate a phase of rapid downward bending, which is followed by a second, slower phase of curvature response that is accompanied by root-tip oscillations similar to those preceding the gravity stimulus. Distinct Brachypodium accessions differ from each other in the frequency and amplitude of their root tip oscillations, as well as in the bending rate following gravity stimulation.

To better capture the genetic basis of variation between accessions for these traits, we developed a mathematical model that recapitulates the evolution of root tip angles from the vertical before, during, and after gravity stimulation. When applied to the 45 accessions under investigation, this model was very effective at recapitulating accession-specific growth behaviors, with best-fit curves correlating well with the observed data (R-square larger than 0.9). These best-fit equations, along with Fast Fourier Transform analysis, allowed estimation of descriptive parameters for these behaviors, including root growth rate, rapid bending rates during phase 1 of gravitropism, and frequency and amplitude of oscillations.

Because each one of these 45 accessions had previously been subjected to genome sequencing, we were able to use genome-wide association studies (GWAS) to identify single nucleotide polymorphisms (SNPs) that are associated with these parameters. SNPs showing significant probabilities of association with several growth parameters including the maximal rate of rapid bending and the amplitude of oscillations, have been identified and are being characterized. Overall, these studies have allowed us to identify linked loci whose contribution to oscillatory root growth behavior are being investigated using a combination of expression analysis, reverse genetics and cell biology approaches.

Overall, these studies may impact our understanding of the molecular mechanisms that govern the regulation of coordinated cell expansion and its roles in growth regulation, plant morphogenesis, and plant organs' growth responses and adaptation to gravity and/or microgravity. In the longer term, our studies may enable us to engineer plants that are better suited to function as key components of bioregenerative life support systems for long-term space exploration missions, or more simply for effective growth and productivity in natural and/or agricultural ecosystems on Earth.

Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Su S-H, Stephens SD, Campbell AM, Levine H, Masson PH. "Using Brachypodium distachyon to Investigate Monocot Plant Adaptation to Spaceflight." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
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